An Elementary Proof of the Schläfli Formula

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Abstract

This paper presents an elementary proof of the Schläfli formula in $\mathbb{E}^3$. By utilizing a tetrahedral decomposition method, we reduce the problem from arbitrary convex polyhedra to the specific case of a tetrahedron. The proof for the tetrahedron is then established through analytic geometry and algebraic calculation using the adjugate matrix.

1 Statement of the Main Result

Let P be an arbitrary convex polyhedron in Euclidean 3-space $\mathbb{E}^3$. Suppose that the positions of its vertices are smooth functions of a time parameter t, and that the polyhedron remains non-degenerate at all times. Let $\mathcal{L}$ be the set of all edges of P. For any edge $l \in \mathcal{L}$, let θ_l denote the interior dihedral angle associated with the edge l. Then, at any instant t, the following equality holds:

$$\sum_{l\in\mathcal{L}} l\frac{d\theta_l}{dt} = 0 \tag{1}$$

2 Proof

2.1 Lemma: Reduction to Tetrahedra

Lemma 1. *If the Schläfli Formula holds for any tetrahedron, then it holds for any arbitrary convex polyhedron.*

Proof. Consider an arbitrary polyhedron P. We perform the following "tetrahedral decomposition":

(i) For each face of the polyhedron, choose a point in its interior. Connect this point to each vertex of the face, dividing the face into several triangles.

(ii) Choose a point in the interior of the polyhedron. Connect this point to all vertices of the polyhedron, as well as to the points chosen in step (i). This divides the polyhedron into a set of tetrahedra, denoted by $\mathcal{S}$.

For each tetrahedron $s \in \mathcal{S}$, let $\mathcal{L}_s$ be the set of its edges. For any $l \in \mathcal{L}_s$, let $\theta_{l,s}$ denote the interior dihedral angle of the tetrahedron s at edge l. By hypothesis, if the formula holds for tetrahedra, we have:

$$\sum_{s\in\mathcal{S}}\sum_{l\in\mathcal{L}_s} l\frac{d\theta_{l,s}}{dt} = 0 \tag{2}$$

Let $\mathcal{L}'_P$ denote the set of all edges in the decomposed polyhedron (including those created by the subdivision). We partition $\mathcal{L}'_P$ into three disjoint sets:

(i) $\mathcal{L}_P$: The set of original edges of the polyhedron P (excluding those from subdivision).

(ii) $\mathcal{L}_{1,P}$: The set of edges lying on the surface (faces) of P generated by the subdivision.

(iii) $\mathcal{L}_{2,P}$: The set of edges lying in the interior of P generated by the subdivision.

Thus, we have the disjoint union:

$$\mathcal{L}'_P = \mathcal{L}_P \sqcup \mathcal{L}_{1,P} \sqcup \mathcal{L}_{2,P} \tag{3}$$

Let Θ_l be the set of all dihedral angles incident to an edge l. We observe the following geometric identities regarding the sum of angles around an edge:

$$\begin{aligned}
\sum_{\theta\in\Theta_l} \theta &\equiv 2\pi, && \forall l \in \mathcal{L}_{2,P} \\
\sum_{\theta\in\Theta_l} \theta &\equiv \pi, && \forall l \in \mathcal{L}_{1,P} \\
\sum_{\theta\in\Theta_l} \theta &\equiv \theta_l, && \forall l \in \mathcal{L}_{P}
\end{aligned}$$

where θ_l in the last equation represents the interior dihedral angle of the original polyhedron P at edge l.

Differentiating both sides with respect to t and multiplying by the edge length l, we obtain:

$$\begin{aligned}
\sum_{\theta\in\Theta_l} l\frac{d\theta}{dt} &\equiv 0, && \forall l \in \mathcal{L}_{2,P} \\
\sum_{\theta\in\Theta_l} l\frac{d\theta}{dt} &\equiv 0, && \forall l \in \mathcal{L}_{1,P} \\
\sum_{\theta\in\Theta_l} l\frac{d\theta}{dt} &\equiv l\frac{d\theta_l}{dt}, && \forall l \in \mathcal{L}_{P}
\end{aligned}$$

Now, rearranging the summation in Equation (2) by grouping terms according to edges l, we derive:

$$\begin{aligned}
0 &= \sum_{s\in\mathcal{S}}\sum_{l\in\mathcal{L}_s} l\frac{d\theta_{l,s}}{dt} = \sum_{l\in\mathcal{L}'_P}\sum_{\theta\in\Theta_l} l\frac{d\theta}{dt} \\
&= \sum_{l\in\mathcal{L}_P}\sum_{\theta\in\Theta_l} l\frac{d\theta}{dt} + \sum_{l\in\mathcal{L}_{1,P}}\sum_{\theta\in\Theta_l} l\frac{d\theta}{dt} + \sum_{l\in\mathcal{L}_{2,P}}\sum_{\theta\in\Theta_l} l\frac{d\theta}{dt} \\
&= \sum_{l\in\mathcal{L}_P} l\frac{d\theta_l}{dt} + 0 + 0
\end{aligned} \tag{4}$$

This completes the proof of the lemma. □

2.2 Proof for the Tetrahedron Case

Proof. As shown in Figure 1, let the origin be O. All variables defined below are functions of time t. Let the position vectors of the vertices be $\vec{a} = \overrightarrow{OA}$, $\vec{b} = \overrightarrow{OB}$, and $\vec{c} = \overrightarrow{OC}$. Let their coordinates be $\vec{a} = (u_1, v_1, w_1)$, $\vec{b} = (u_2, v_2, w_2)$, and $\vec{c} = (u_3, v_3, w_3)$.

The vectors $\vec{a}\times\vec{b}$, $\vec{b}\times\vec{c}$, and $\vec{c}\times\vec{a}$ are normal vectors to the faces pointing towards the *interior* of the tetrahedron. Let $\vec{d} = \overrightarrow{CA}\times\overrightarrow{CB} = (\vec{a}-\vec{c})\times(\vec{b}-\vec{c}) = \vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}$. This vector $\vec{d}$ is the normal vector pointing towards the *exterior* of the fourth face.

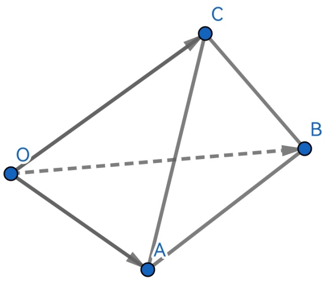

Figure 1: Notation for the tetrahedron.

Let $\vec{b} \times \vec{c} = (x_1, y_1, z_1)$, $\vec{c} \times \vec{a} = (x_2, y_2, z_2)$, and $\vec{a} \times \vec{b} = (x_3, y_3, z_3)$. Then $\vec{d} = (x_1 + x_2 + x_3, y_1 + y_2 + y_3, z_1 + z_2 + z_3)$.

Define the matrix A and its matrix of cofactors (specifically, the transpose of the matrix of cofactors, known as the adjugate matrix) A^* as:

$$A = \begin{bmatrix} u_1 & v_1 & w_1 \\ u_2 & v_2 & w_2 \\ u_3 & v_3 & w_3 \end{bmatrix}, \quad A^* = \begin{bmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{bmatrix}$$

Based on our construction, we have the property:

$$AA^* = A^*A = (\det A)I$$

which implies A^* is indeed the adjugate matrix of A. It follows easily that $\det A^* = (\det A)^2$. Consequently:

$$((\det A)A)A^* = A^*((\det A)A) = (\det A)^2 I$$

This implies that $(\det A)A$ is the adjugate matrix of A^* (a fact used in later calculations).

Using the relation $l \cdot \frac{d\theta}{dt} = -l \cdot \frac{d(\cos\theta)}{dt} \cdot \frac{1}{\sin\theta}$, it suffices to prove:

$$\sum_{l \in \mathcal{L}} l \cdot \frac{d(\cos\theta)}{dt} \cdot \frac{1}{\sin\theta} = 0 \tag{5}$$

Due to the symmetry of the variables, we only need to calculate terms for two representative types of edges:

Case 1: Edge AB (Representative of edges on the base triangle)

$$l_{AB} = \sqrt{(u_1 - u_2)^2 + (v_1 - v_2)^2 + (w_1 - w_2)^2}$$

$$\frac{1}{\sin\theta_{l_{AB}}} = \frac{|\vec{d}||\vec{a} \times \vec{b}|}{|\vec{d} \times (\vec{a} \times \vec{b})|} = \frac{|\vec{d}|\sqrt{x_3^2 + y_3^2 + z_3^2}}{M_0}$$

where M_0 is calculated as:

$$\begin{aligned} M_0 &= |(x_3, y_3, z_3) \times (x_1 + x_2 + x_3, y_1 + y_2 + y_3, z_1 + z_2 + z_3)| \\ &= \sqrt{\begin{vmatrix} y_3 & z_3 \\ y_1 + y_2 + y_3 & z_1 + z_2 + z_3 \end{vmatrix}^2 + \begin{vmatrix} z_3 & x_3 \\ z_1 + z_2 + z_3 & x_1 + x_2 + x_3 \end{vmatrix}^2 + \begin{vmatrix} x_3 & y_3 \\ x_1 + x_2 + x_3 & y_1 + y_2 + y_3 \end{vmatrix}^2} \\ &= \sqrt{(\begin{vmatrix} y_3 & z_3 \\ y_1 & z_1 \end{vmatrix} + \begin{vmatrix} y_3 & z_3 \\ y_2 & z_2 \end{vmatrix})^2 + (\begin{vmatrix} z_3 & x_3 \\ z_1 & x_1 \end{vmatrix} + \begin{vmatrix} z_3 & x_3 \\ z_2 & x_2 \end{vmatrix})^2 + (\begin{vmatrix} x_3 & y_3 \\ x_1 & y_1 \end{vmatrix} + \begin{vmatrix} x_3 & y_3 \\ x_2 & y_2 \end{vmatrix})^2} \\ &= |\det A|\sqrt{(u_2 - u_1)^2 + (v_2 - v_1)^2 + (w_2 - w_1)^2} \end{aligned}$$

For the cosine of the dihedral angle:

$$\cos\theta_{l_{AB}} = \frac{\vec{d}\cdot(\vec{a}\times\vec{b})}{|\vec{d}||\vec{a}\times\vec{b}|} = \frac{x_3(x_1+x_2+x_3)+y_3(y_1+y_2+y_3)+z_3(z_1+z_2+z_3)}{|\vec{d}|\sqrt{x_3^2+y_3^2+z_3^2}}$$

Note that the normal vectors point in opposite directions (one inward, one outward), so no negative sign is needed. Let $S_3 = x_3(x_1+x_2+x_3)+y_3(y_1+y_2+y_3)+z_3(z_1+z_2+z_3)$. Differentiating yields:

$$(\cos\theta_{l_{AB}})' = \frac{S_3'|\vec{d}|\sqrt{x_3^2+y_3^2+z_3^2} - \frac{x_3x_3'+y_3y_3'+z_3z_3'}{\sqrt{x_3^2+y_3^2+z_3^2}}S_3|\vec{d}| - \sqrt{x_3^2+y_3^2+z_3^2}S_3|\vec{d}|'}{x_3^2+y_3^2+z_3^2}$$

Thus:

$$l_{AB}\frac{(\cos\theta_{l_{AB}})'}{\sin\theta_{l_{AB}}} = \frac{1}{|\det A|}\left(S_3' - \frac{x_3x_3'+y_3y_3'+z_3z_3'}{x_3^2+y_3^2+z_3^2}S_3 - S_3\frac{|\vec{d}|'}{|\vec{d}|}\right) \tag{6}$$

Case 2: Edge OA (Representative of edges incident to origin)

$$l_{OA} = \sqrt{u_1^2+v_1^2+w_1^2}$$

$$\frac{1}{\sin\theta_{l_{OA}}} = \frac{|\vec{c}\times\vec{a}||\vec{a}\times\vec{b}|}{|(\vec{c}\times\vec{a})\times(\vec{a}\times\vec{b})|} = \frac{\sqrt{x_2^2+y_2^2+z_2^2}\sqrt{x_3^2+y_3^2+z_3^2}}{M_1}$$

where $M_1 = |\det A|\sqrt{u_1^2+v_1^2+w_1^2}$. For the cosine:

$$\cos\theta_{l_{OA}} = \frac{(\vec{c}\times\vec{a})\cdot(\vec{a}\times\vec{b})}{|\vec{c}\times\vec{a}||\vec{a}\times\vec{b}|} = -\frac{x_2x_3+y_2y_3+z_2z_3}{\sqrt{x_2^2+y_2^2+z_2^2}\sqrt{x_3^2+y_3^2+z_3^2}}$$

Here, both normal vectors point inward, so we introduce a negative sign. Let $T_1 = x_2x_3 + y_2y_3 + z_2z_3$. Differentiating and simplifying yields:

$$l_{OA}\frac{(\cos\theta_{l_{OA}})'}{\sin\theta_{l_{OA}}} = -\frac{1}{|\det A|}\left[T_1' - \left(\frac{x_2x_2'+y_2y_2'+z_2z_2'}{x_2^2+y_2^2+z_2^2} + \frac{x_3x_3'+y_3y_3'+z_3z_3'}{x_3^2+y_3^2+z_3^2}\right)T_1\right] \tag{7}$$

Summation Let $Q_3 = \frac{x_3x_3'+y_3y_3'+z_3z_3'}{x_3^2+y_3^2+z_3^2}$. We can rewrite equations (6) and (7) as:

$$l_{AB}\frac{(\cos\theta_{l_{AB}})'}{\sin\theta_{l_{AB}}} = \frac{1}{|\det A|}\left(S_3' - Q_3S_3 - S_3\frac{|\vec{d}'|}{|\vec{d}|}\right) \tag{8}$$

$$l_{OA}\frac{(\cos\theta_{l_{OA}})'}{\sin\theta_{l_{OA}}} = -\frac{1}{|\det A|}\left[T_1' - (Q_2+Q_3)T_1\right] \tag{9}$$

We sum these expressions cyclically (denoted by $\sum_{\text{cyc}}$):

$$
\begin{aligned}
|\det A| \sum_{l\in\mathcal{L}} l\frac{(\cos\theta_l)'}{\sin\theta_l} &= |\det A| \left(\sum_{\text{cyc}} l_{AB}\frac{(\cos\theta_{l_{AB}})'}{\sin\theta_{l_{AB}}} + \sum_{\text{cyc}} l_{OA}\frac{(\cos\theta_{l_{OA}})'}{\sin\theta_{l_{OA}}} \right) \\
&= \sum_{\text{cyc}} (S_3' - Q_3S_3 - S_3\frac{|\vec{d}|'}{|\vec{d}|}) - \sum_{\text{cyc}} (T_1' - (Q_2+Q_3)T_1) \\
&= \sum_{\text{cyc}} S_1' + \sum_{\text{cyc}} Q_3(T_1+T_2-S_3) - \sum_{\text{cyc}} T_1' - \frac{|\vec{d}|'}{|\vec{d}|}\sum_{\text{cyc}} S_3 \\
&= \sum_{\text{cyc}} S_1' - \sum_{\text{cyc}} Q_3(x_3^2+y_3^2+z_3^2) - \sum_{\text{cyc}} T_1' - \frac{|\vec{d}|'}{|\vec{d}|}|\vec{d}|^2 \\
&= \sum_{\text{cyc}} S_1' - \sum_{\text{cyc}} (x_3x_3'+y_3y_3'+z_3z_3') - \sum_{\text{cyc}} T_1' - |\vec{d}||\vec{d}|' \\
&= \sum_{\text{cyc}} S_1' - \frac{1}{2}\sum_{\text{cyc}} (x_1^2+y_1^2+z_1^2+2T_1)' - \frac{1}{2}(|\vec{d}|^2)' \\
&= (\sum_{\text{cyc}} (x_1+x_2+x_3)^2)' - \frac{1}{2}(\sum_{\text{cyc}} (x_1+x_2+x_3)^2)' - \frac{1}{2}(\sum_{\text{cyc}} (x_1+x_2+x_3)^2)' \\
&= 0
\end{aligned}
$$

Since $|\det A| \neq 0$ (due to the non-degenerate assumption), the proof is complete. $\square$

3 Conclusion

1. The proof provided here is computational in nature and does not explicitly reveal the geometric intuition behind the Schläfli Formula. However, the precise cancellation of terms during the calculation reveals an intrinsic algebraic beauty. A deeper investigation into these algebraic expressions might yield further geometric insights.
2. Every step of this computation is independent of the assumption that the dimension is three. Furthermore, the lemma regarding barycentric subdivision is dimension-independent. Therefore, this method of proof can be readily generalized to n-dimensional Euclidean space.